Making our world more productive



Proposal for LMO Installation 1X20 KL at JECRC Medical College, Hospital & Jaipur (RJ)

ABOUT US

We, at Linde, India, are privileged to have this opportunity to serve you. Linde India Limited, has had a well-established presence in India since 1935, and has grown into one of the leading gases and engineering companies in India, with one of the most comprehensive industrial, specialty and medical gases product portfolios, along with the largest direct sales and distribution networks across the nation, serving a variety of industry segments.

Keeping our customers at the center of everything we do, we aim at driving efficiency into their processes, reducing energy costs, safeguarding people and protecting the environment.

Linde plc is world's leading player in industrial gases, with sales in 2018 of $ 28 billion. The company employs approximately 80,000 people and serves customers in more than 100 countries around the world, delivering innovative and sustainable solutions and creating long-term value for all stakeholders. Our mission is to build a more productive world by providing products, technologies and services that help our customers improve their economic and environmental performance in a connected world. Linde India Ltd., a wholly owned subsidiary of Linde plc with its headquarters in Kolkata, is the leading industrial gases supplier in India, with more than 41 operating locations, a national distribution network, and an impressive list of clients.

With access to vast resource pool, wide network of plants, mobilization capability, and on-time product delivery through a national distribution network, Linde provides various supply systems, end-to-end services along with supply of various industrial, medical and specialty gases to meet & exceed customer needs.

Strategically located cylinder bottling units across the country and a strong fleet of cryogenic bulk tankers enables Linde India to be a leader in complete range of Industrial gas supplies in the country.

For more information about the company and its products and services, please visit www.linde.com.

INTRODUCTION- LIQUID MEDICAL OXYGEN (LMO)

Virtually every hospital needs oxygen and the most economical way to hold large quantity is in a liquid form. Oxygen liquefies at –183 deg.C and to keep it this cold a specific type of container is needed. This special vessel is called a Vacuum Insulated Evaporator or VIE.

There are additional advantages to keeping oxygen in liquid form, as well as the cost considerations. It requires minimal maintenance and is easy to access and use. In most hospitals, all you see of the system is the tap in the wall next to a bed. The

VIE tanks are located outside the hospital buildings and LINDE ensures they are regularly re-supplied with oxygen.

Oxygen is drawn off as needed, passed through a vaporizer, and turned into gas. A control station regulates the flow of the gas from the VIE before it enters the hospital network pipework. This is an essential safety feature, and an area where LINDE expertise and experience is second to none.

LINDE'S TECHNICAL & SAFETY FEATURES

Our impeccable record for safety is a by adopting the global safety standards over and above the Indian statutory guidelines.

LINDE Group's global safety standards, we will incorporate all such high level safety standards in the Liquid medical oxygen storage system in your hospital for further enhancement of safe practices in handling of medical oxygen supply. The features and benefits of such initiative is explained below: -

Mandatory alarm signals:

- Primary vessel contents are low
- Primary vessel pressure is low (Back up supply is in operation).
- Hospital pipeline pressure is low

Equipment requirements:

- Dual parallel regulators in customer pipeline & open/online. No by-pass systems to pressure regulators.
- Regulators to be in the set position.
- Three way valves on alarm switches.
- Thermal type relief valve must not be fitted as pipeline pressure relief valve, must be a certified performance Safety valve

System Components

- TWIN LINE PRESSURE REGULATOR
  - Ensures uninterrupted supply of medical oxygen to hospital all the time at constant pressure, 4.2 Bar as per international practice. One regulator is set at 4.2 Bar and other one set at 3.8 bar.
  - Facilitates for repair / replacement if required at any time without shut down of oxygen supply to hospital.
- ALARM - VIE CONTENT LOW LEVEL
  - This alarm gives an audio - visual indication to hospital staff alerting them time of re-ordering level of liquid medical oxygen, thus eliminates any chances for VIE to run out of stock.
- ALARM - VIE LOW PRESSURE & BACK UP MANIFOLD IN OPERATION
  - This alarm gives an audio - visual indication to hospital staff in case VIE pressure goes below set pressure and back up manifold in operation. This alerts hospital staff about any low-pressure situation in VIE and immediate need for inspection / checking etc to rectify the fault.
- ALARM - HEALTH CARE FACILITY PIPELINE PRESSURE IS LOW
  - This alarm gives an audio - visual indication when hospital supply pipeline pressure fall below the set pressure, thus alerting the necessity for inspection / checking and immediate need for rectification of the fault.
- ALARM - BACK UP SUPPLY CONTENT LOW
  - This alarm gives an audio - visual indication when pressure at one side of manifold has reached a level where cylinders on that side need to be replaced.
  - Two separate alarms for cylinder content status of right bank and left bank is provided in this system and this makes the system more reliable.

TWIN PIPELINE SAFETY RELIEF VALVE (SRV) SYSTEM (SRV-1(2nos) and SRV-2 (2nos)

This system has been provided for over pressure protection of oxygen supply pipeline to Hospital. 3-Way diverter valve (DV-1 and DV-2 as indicated in P&ID) ensures that at least one safety valve is always online and eliminates any chances for over pressurization of Hospital pipe line at any instant of time.

3-WAY GAUGE COCK VALVE

The 3-Way gauge cock valve is for isolating the line pressure switch with manual vent of locked pressure and is built in 3-Way route and eliminates false low pressure alarm indication

ENSURE QUALITY SERVICE: WHY LINDE?

- Vacuum insulated vessels which ensures that the losses are minimal.
- Required vessel capacity as per Design. The unique feature of LINDE is the **National Scheduling Center (NSC)** to suit the needs of each customer. Vessel size is determined based on the demand pattern, demand fluctuation, maximum draw-off rates, duration and distance from the Liquid Medical Oxygen production centre.
- Liquid Medical Oxygen multi production centers across the country and very high storage capacity in all the regions to facilitate cross transfer, if needed. In addition more than 245 VITT's (Bulk Tankers) are spread across the country for delivery.
- Robust logistics to ensure supply. Liquid Medical Oxygen supplies continue even though there are civil disturbances or transport strikes.
- Product quality better than the IP specified purity. Every batch monitored by Drug Inspectorate approved specialists.
- All LINDE Liquid Medical Oxygen installations are equipped with set of alarms and sensors to monitor Vessel pressure and volumes
- Customer training is given by our engineers on the alarm systems, re-ordering and emergency procedures. LINDE's Medical Installation Engineering Team provides round the clock maintenance services at customer premises.

ADVANTAGES OF LINDE LIQUID MEDICAL OXYGEN VESSEL:

· Only company in India having all India network of well-trained service engineers thereby giving excellent service backup.

· All India network of Liquid Medical Oxygen supply.

· Only company with dedicated National Scheduling Center for all Liquid Supplies apart from Customer Care Center.

· Telemetry system install at the vessel for the real time data monitor of stock and pressure recorded at national scheduling Center

On-Line Tracking of Vehicles with real time tracking of Speed Monitoring , Vehicle duty & track route violation if any. The NSC has a capability to track down individual customer wise supply time, speed limit etc. for enabling the product to reach the

PROPOSED SCHEME

- This project is CAPAX based project.
- It is proposed to install a Vacuum Insulated Storage Vessel in close proximity of the manifold. Liquid Medical Oxygen shall be carried from our stock at local office, duly tested as per IP standards, and will be decanted into the vessel from time to time.
- Linde will Install the Liquid Medical Oxygen at their own cast.
- Hospital Need to provide the Space as per SMPV Rule 2016 for the installation of the vessel.
- The regular Maintenance of the vessel and regular supply to maintain the sufficient stock 24*7 as per contracted volume will be the Linde Responsibility.

ANNEXURE 1

SCOPE OF WORK

Seller shall supply and install the "Equipment" (as detailed in Annexure 2) at the Buyer's Works. If any new or modified local, state or central law, rule or regulation requires modification to the Equipment or site, or requires that any additional permits, licenses or rights be obtained, any costs associated with performing such modifications or obtaining such permits, licenses or rights will be borne by Buyer.

Sl.	Work Item	Seller	Buyer
1	Design, Supply, Assemble, erect, install, pressure test & commission the supply system	√	-
2	Provide the basic engineering drawings for the Equipment	√	-
3	Product Supply as per Annexure 2	√	-
4	Support for the **PESO** licensing and assistance for installation	√	-
5	Train selected employees of buyer on handling of the supply system & safety features	√	-
6	Man the installation for **1-2** days during startup, as per the requirement	√	-

7	Buyer to provide an open area of approximately 9 **m x 17 m** for the above Liquid Medical Oxygen vessel Installation (as specified by PESO)	-	√
8	All the Civil Drawings pertaining to the Installation will be provided by Seller on receipt of the Order	√	-
9	The Statutory Fees levied by PESO, Nagpur towards getting the Installation License for Storage shall have to be borne by Buyer	-	√
10	**All necessary Civil/ Mechanical/ Electrical Work, mentioned below, for effective commissioning of the Equipment shall be done by Buyer as per Seller guidelines**	-	√
	a)All Civil Foundation LMO Storage Vessel, Hardstand, Vaporizer, etc.) & Structural work (if any)	-	√
	b)Fencing of the Location (if any)	-	√
	c)Digging of Trench for laying the Pipeline (if required)	-	√
	d)Foundation of Structural Support for Pipeline (if any)	-	√
	e)Other Punch List Items (earth pits, fire extinguishers, industrial lighting – flameproof, etc. if any)	-	√
	f) Visit to PESO – Nagpur/local office for license and endorsement purpose by one of Buyer's representative for the License. Buyer shall make arrangements for local PESO officers for inspection and endorsement of the license.	√-	
	g) Distribution pipeline from the Equipment to and inside the Hospital shall be Buyer's responsibility.	-	√
	h) All electrical jobs like area lighting, earthing, power sockets, and drawing power cable to the control panel shall be Buyer's responsibility per guidelines provided by Seller.	-	√
11	**Facilities required during the time of erection & commissioning of installation**	-	√
	a) Free access to Buyer's Works for Seller and its representatives at all times. All utilities like Power, Water etc. and secured storage space for keeping Seller's material to be used during Installation shall be provided by Buyer.	-	√
	b) Permanent provision for Water shall be provided by Buyer at the Installation Site.	-	√

	c) Buyer shall provide tested and certified Crane as specified by the Seller, for unloading and erection of the Equipment.	-	√
	d) Provide the required gas and consumable for Pressure Testing of the Equipment	-	√
12	Equipment Maintenance	√	-
	Seller shall maintain Equipment.	√	-
	The Maintenance Contract shall cover planned (regular) maintenance of the Installation including:	√	-
	i. Annual Pressure Testing (gas & consumable to Buyer account),	√	-
	ii. Replacement of Defective (either due to material or Workmanship defect) and worn out parts.	√	-
13	Any other approvals from other statutory bodies	-	√
14	Unplanned maintenance due to any Installation defect arising out of mishandling, pilferage, damage, etc., on actual chargeable basis.	√	-
15	Housekeeping & Maintenance of the "Installation Site" shall be done by BUYER	-	√

Hospital Scope Of work:-

- Open, clear space as per requirement off CCOE (PESO) for installation of LMO vessel and AV Coil, with clear space for tanker approach.
- To make foundation for LMO Vessel and AV Coil as per LINDE specifications and necessary fencing and safety equipment's as per CCOE guidelines (Details will be provided by LINDE)
- Secured area for unloading and storage of LMO vessel and AV Coil during the installation phase.
- Necessary work permits / entry gate passes required by LINDE/ LINDE sub-contractors for installation of the vessel.
- Medical Gas Pipeline from SMPV Area to Manifold room will be the Hospital responsibility
- Security of the SMPV Area 24*7
- Any other statutory approval if required by Local Body/Govt. agencies.

CCOE [PESO] GUIDELINES

1. 6Ft High fencing of Industrial gauge
2. Hard stand for the VIE
3. Anti Crash barrier
4. Fire Buckets
5. 2 Fire Extinguishers
6. No moving parts or machinery in the vicinity
7. No underlying cables in the premises
8. No overhanging cables in the vicinity
9. No over looking windows or balconies
10. No residential accommodations with in premises.

ANNEXURE 2

COMMERCIAL OFFER

Supply & Installation of Liquid Medical Oxygen -20KL Vessel

Description	Liquid Medical Oxygen as per IP
Product Price	13.21sm3
Delivery Charges	4.79-sm3
Faciality charges	24000/-per month
One time installation charges	200000/-

- All above charges are exclusive of all applicable taxes, Octroi, cess and duties and in the event is or becomes applicable, shall be paid extra by Buyer over and above this amount.
- Gas Consumption Pattern

The above pricing is derived based on the current Gas consumption pattern communicated.

Medical Oxygen in Bulk : 20000 M3 per Month
Equipment details: -

- 20 KL Vessel
- 2*500 nm3/Hr. Vaporizer
- Telemetry- 1 Nos.
- O&ED Alarm Panel
- Asset ownership-Linde India Limited

- Price Escalation

The current pricing is based on current Power, Diesel & Inflation rate; an escalation will be applicable on Basic & Delivery Charges.

Diesel:
For every 1 rupee increase in diesel prices, there shall be a corresponding increase in the price of liquid medical oxygen by 10 paisa per cubic meter. Diesel price at the time of signing agreement will be taken as base diesel price.

Power:
For every 1 paisa increase in power prices at BOCI'S supplying unit, there shall be a corresponding increase in the price of liquid medical oxygen by 1 paisa per cubic meter. Last quarter power price will be taken as base power base at the time of signing agreement.

OR

- Price escalation – 7% Year on Year Escalation on Facility Charges.

- <u>Mode of Delivery</u>

Product shall be supplied by Linde on **Weighment Basis** through Linde own Transport Tanker (Vacuum Insulated Transport Tanker, VITT) and the product will be decanted in the Storage Tank (Vacuum Insulated Storage Tanker, VIST) installed at Buyer premises.

The **VITT** will be weighed at any approved Weight Bridge before and after decanting Liquid into the VIST. The difference in the weight will be the quantity delivered to the buyer.

- Payment Terms

  Terms of payment will be net Thirty(30) days from Seller's invoice date. Notwithstanding anything contained herein above, in the event of Buyer failing to make timely payments and the amount remaining unpaid even after the credit period, overdue amount shall attract an interest at the rate of 2% per month till the date of actual payment. However, in addition to claim interest for the delayed payment Seller may at its discretion limit or stop further supply of the Product, and/or terminate this Agreement and remove the Equipment from the Buyer's Works with or without notice of legal process, and/or suspend Seller's performance under this Agreement and / or terminate this Agreement.

- Duration of Contract

  Seller shall enter into a 05 **(five) years Contract** (Minimum Supply Period with Buyer for supply of **Liquid Medical Oxygen** which will be renewed after the agreed Contract Period subject to mutual agreement between Seller and Buyer.

- Validity

  The above terms & conditions of our Proposal shall be valid for acceptance for a period of 3 **(three) month** from the date of the offer and thereafter subject to our confirmation.

- Backup

  Back up cylinders to be provided by the hospital and gas manifold has to be in working condition.

Estimated time for delivery and completion

* Procurement & Supply of materials at job site within 8 to 12 weeks from the date of receipt of your formal order along with the advance.

* Fabrication, installation, testing, commissioning and handing over will take within 8 to 12 weeks' time, depending upon the site availability, CCOE approvals.

* Hospital should arrange at least48 Hours backup of Liquid Medical Oxygen by the hospital to avoid any emergency

We assure you that we maintain very high standards of quality and service and are confident of meeting your expectations.

Thanking You
Hardeep Singh
9878219520
FSR-LUDHIANA
Linde India Limited



Registered Office
Linde India Limited
Oxygen House, P43 Taratala Road, Kolkata 700 088, India
Phone +91 33 6602-1600; Fax +91 33 2401 4206; customercare.lg.in@linde.com; www.linde.in;